

Section A

Data Fundamentals

1. A database is an organized collection of data that can be easily stored, accessed, and managed. E.g. A school system that stores names, grades and attendance.

2. Data warehouse is a system that stores large amounts of historical from different sources for reporting and analysis. E.g. A company collects sales data from all its branches over many years and stores it in one place for analysis.

3) Data lake is a storage system that holds raw data in its original format until it is needed. E.g. A company stores all kinds of data - videos, E

4. Data mart is a smaller, focused part of a data warehouse designed for a specific department or purpose.

E.g. The finance department has its own smaller database with only financial data (budgets, expense revenue) taken from the main data warehouse.

5. Structured Data is a neatly organised data in a fixed format (rows and columns)

E.g. A table of students with columns like name, Age and marks

* Semi-structured Data it is a partly

that is not in a strict table but still has some structure (uses tags or labels).
E.g (name: John, age: 20).

* Unstructured data is a data that is not organized at all with no specific format or structure.

E.g. A video file or a voice recording

6) Data modelling is the process of organizing and structuring data so it can be stored, managed, and used efficiently in a system. it defines what data is needed, how it is related, and how it should be ~~transfer~~ formatted

Why it is important:

- * It helps ensure the system stores the right data in the right way.
- * Improves data quality and consistency
- * makes it easier to design databases and applications
- * Helps different teams understand the data clearly
- * Reduces errors and duplication

Types of data modelling

* Conceptual data model → This is a high-level model that shows what data exists and the relationships between them. e.g: Showing that a customer can place many orders without going into technical details.

* Logical data model → This adds more details by defining attributes, keys and relationships, but still does not depend on a specific database system.

7) Data mining is the process of analyzing large amounts of data to discover useful patterns, trends and relationships that are not obvious at first.

In simple terms: it's how businesses turn raw data into meaningful insights for decision-making.

How data mining is used in business

Businesses use data mining to:

- * understand customers (buying habits, preferences)
- * Improve marketing (target the right audience)
- * Increase sales (recommend products)
- * Reduce risks (detect fraud or unusual activity)
- * make better decisions based on real data, not guesses

Real-world example of data mining

A. Amazon

Amazon uses data mining to analyze:

- * what you search
- * what you buy
- * what other customers

B. Netflix

Netflix analyzes:

- * What you watch
- * how long you watch
- * What you skip

8. Database Management system (DBMS): it is a software that allows users to create, store, manage, and retrieve data in a structured and efficient way.

Popular DBMS Platforms

A. MySQL

- * open-source relational DBMS, easy to use and widely supported.
- * use case: web applications (e.g. website, blogs, e-commerce).

B. PostgreSQL

- * Powerful open-source DBMS with advanced features and strong data integrity.
- * use case: complex system (e.g. financial systems, analytics platforms).

C. Oracle Database

- * Commercial DBMS known for high performance, security and scalability.
- * use case: large enterprise (e.g. banking, large organization).

D. MongoDB

- * No SQL DBMS that stores data in flexible, JSON-like documents.

- * Use case: Applications with unstructured data, e.g. social media, real time apps.

9. ETL (Extract, Transform, Load) is a process used to collect, clean and store data for analysis.

Steps:

- * Extract: Get data from different sources (databases, files).

- * Transform: Clean and convert the data into a useful format.

- * Load: Store the processed data in a system like a data warehouse.

Importance:

ETL ensures data is clean, consistent, and ready for analysis, helping businesses make better decisions.

10. Common data storage formats used in analytics

A. CSV (comma-separated values)

- * Simple text format where data is separated by commas.

- * Use when: ~~working with APIs~~ data is small, simple and needs to be easily shared or opened in tools like Excel.

B. JSON (JavaScript Object Notation)

- * stores data in key-value pairs, often nested.
- * use when: working with APIs or semi-structured data.

C. Parquet

- * columnar storage format optimized for big data.
- * use when: handling large datasets and doing fast analytics (better performance and compression).

D. AVRO

- * Row-based format with a defined schema.
- * use when: data needs to be exchanged between systems and schemas may change over time.

Section B: Artificial Intelligence & Machine Learning

11. Artificial Intelligence (AI):

AI is the ability of machines or computer systems to perform tasks that normally require human intelligence, such as learning, reasoning and decision-making.

A. Narrow AI (Weak AI)

- * Designed to perform a specific task only.
 - * Cannot operate beyond its programmed function.
- ~~Example:~~ Siri - answers questions and performs commands but only within its scope.

B. General AI (strong AI)

- * Aims to perform any intellectual task like a human
- * can learn, reason and adapt across different areas

Example: no true real world example yet (still theoretical), but fictional systems like Jarvis represent this idea.

12. Machine Learning (ML) is a branch of Artificial Intelligence where computers learn patterns from data and make decisions or predictions without being explicitly programmed for every rule.

Difference from Traditional Programming

A. Traditional Programming:

- * You give the computer rule (code) + data $\Rightarrow$ output
- Example: you write exact steps to calculate something.

B. ~~use~~ Machine Learning:

- * You give computer data + expected output.
- $\Rightarrow$ it learns the rules (model)

(i) Traditional = human writes rules

(ii) machine learning = machine learns rules from data.

13. Deep learning is a subset of machine learning that uses complex models called neural networks to learn from large amounts of data - especially for tasks like images, speech and language.

Relationship

- * Artificial Intelligence (AI) = the broad field (machines acting smart)
- * machine learning = a part of AI (learning from data)
- * Deep Learning = a part of ML (uses neural networks)

A. Neural networks

neural networks are models inspired by the human brain.

- * Made of layers of nodes
- * Each neuron processes input and passes it forward
- * The network learns by adjusting weights based on errors.

14. Natural Language Processing (NLP) is a field of Artificial Intelligence that helps computers understand, interpret and respond to human language (text or speech).

Real-life Applications

a) Voice assistants

- * e.g., Google assistant, Siri
- * Used in various applications like customer support, recommendation systems, etc.

B. Chat apps & autocorrect

* e.g. whatsapp

* suggest words, fix spelling and predict text

C. Translation tools

* e.g. Google Translate

* convert text from one language to another

15. A Large language model (LLM) is a type of Artificial Intelligence system trained on massive amounts of text data to understand and generate human-like language.

Difference from a standard ML model

* Standard machine learning model: built for specific task (e.g., predicting prices, classifying emails) using structured or limited data.

* LLM: trained on huge text datasets and designed mainly to understand and generate natural language.

16. Generative AI is type of AI that creates new content like text, images or code by learning patterns from large amounts of data.

Example of tools:

* ChatGPT - generate text, answers questions, and helps with writing and learning.

* GitHub Copilot - Helps write and suggest code for programmers

* Midjourney - creates images from text description

17. A training dataset is data used to teach an AI model how to recognize patterns and make predictions

Why quality matters

Good - Quality data helps the AI ^{learn} ~~learn~~ correct and accurate patterns.

What goes wrong with bad or biased data.

* Wrong or inaccurate results

* Unfair or biased decisions

* Poor performance in real situation

* It can repeat and even amplify bias in the data.

18. Computer vision is a field of AI that allows computers to see and understand images or videos.

How machines see images?

They don't see like humans. Images are turned into numbers (pixels). AI models - often neural networks - analyze patterns like edges, shapes, colours and makes sense of what's in the image

Two real-world use cases:

- * Face recognition in phone unlocking and security systems
- * Self-driving cars that detect pedestrianly traffic signs, and other vehicles

Section C: APIs, Tools & Developer Concepts

19. API (Application Programming Interface) is a set of rules that allows different software systems to communicate with each other.

Simple analogy:

Think of an API like a waiter in a restaurant. You (the user/app) give your order to the waiter (API), the kitchen (system/server) prepares the food and the waiter brings back your meal. You don't need to know how the kitchen works.

Why APIs are important

- * They let different apps and systems share data and features easily
- * They save time by allowing developers to reuse existing services instead of building everything from scratch.

* They enable modern services like payments, maps, logins and cloud apps to work together smoothly.

20. AN API key is a ~~secret~~ secret code used to identify and authenticate an app when it uses an API.

Purpose:

control access and track usage.

How to protect it

keep it private, use environment variables and never share or upload it publicly.

if exposed:

others can misuse it, cause costs, steal data or get your account blocked.

21. A REST API is a way for systems to communicate over the web using standard rules (HTTP). it is simple, scalable and widely used.

REST stands for:

Representational State Transfer

Main HTTP methods:

- * GET - retrieves data from a server
- * POST - sends new data to a server (create)
- * DELETE - removes data from a server.

22) JSON (JavaScript Object Notation) is a lightweight format used to store and exchange data between systems.

Why it's used in APIs?

- * Easy for humans to read and write
- * Easy for machines to parse and generate
- * Works across different programming languages

Example of JSON:

```
{  
  "name": "John",  
  "age": 25,  
  "city": "Pune"
```

23) Prompt engineering is the skill of writing clear, specific instructions to get better results from AI tools like ChatGPT or Claude.

Why it's important?

- * Gives more accurate and useful answers
- * Saves time by reducing vague or wrong outputs.

Example:

1. Prompt: Explain AI

2. Better: Explain AI in simple terms with 2 real-life examples

2. poor! write something about business

Better! write a short paragraph about how small businesses use social media to grow²⁰

24) A map function applies a function to each item in a list and returns a new ~~list~~ list with the results.

What it does?

Transform data without changing the original list.

Example:

```
numbers = [1, 2, 3]
results = list(map(lambda x: x*2, numbers))
Print(results) # [2, 4, 6]
```

25. Cloud computing is using the internet to access servers, storage and software instead of your local computer.

models:

* IaaS (Infrastructure as a service): Rent servers & storage → Example: Amazon web services

* PaaS (Platform as a service): platform to build apps → example: Google App Engine

* SaaS (Software as a service):

Ready-to use apps → example: Google

26. A version control system (VCS) tracks changes to code/files over time.

Git:

A popular VCS used to manage code, collaborate and keep history of changes.

Why it's used:

- * Easy collaboration
- * Track and revert changes
- * Work on projects safely

Terms:

- * Commit - saving a snapshot of changes
- * Branch - a separate version to work on without affecting the main code

27. A Jupyter Notebook is a tool where you can write and run code, see results and add notes in one place.

Language

most commonly used with Python

Why it's popular

- * combines code, output and explanations
- * Easy for data analysis and visualization
- * Great for experimenting and sharing work

28. Python is a high-level, easy to-read programming language.

Why it's popular for data & AI

- * Simple syntax (easy to learn)
- * Huge community and libraries
- * Strong support for data analysis and machine learning

Common libraries

- * Numpy - numerical computing (array math)
- * Pandas - data manipulation (table, dataframes)
- * Matplotlib - data visualization (chart/graph)
- * Scikit-learn - machine learning tools

~~29~~

Section D: Data Ethics, Governance & Privacy

29. Data Privacy is the protection of personal information and how it is collected, used, and shared.

Why it's important:

- * Protects people's personal data

- * Prevents misuse (eg identity theft)
- * Builds trust between users and companies

General Data Protection Regulation (GDPR):

A strict data protection law that controls how personal data is handled.

Where it applies:

- * All countries in the European Union
- * Also applies to any company worldwide that handles data of EU residents

30. Data ethics is about using data in a way that is fair, responsible, transparent and respectful of people's rights. It guides how data is collected, stored, shared and used so that it doesn't harm individuals or society.

Examples of unethical use of data or AI:

A. Using Personal Data without consent

It violates privacy and trust. People lose control over their own information and may be exposed to risks.

B. Biased AI decision making

Example: An AI system used for hiring favors certain races or genders because it was trained on biased data.

Why it's a problem: It leads to unfair discrimination and reinforces existing inequalities instead of making objective decisions.

2. Spreading misinformation using AI

Example: AI-generated fake news or deepfake videos that make people believe false information.

Why it's a problem: It can mislead the public, damage reputations, and even influence elections or social stability.

31. Data governance is the management of data to ensure it is accurate, secure, and used properly within an organization.

Key components

- * Policies and rules: Guidelines for how data is used and managed
- * Data quality: Ensures data is correct and reliable
- * Roles and ownership: Defines who is responsible for data
- * Security and privacy: Protects data from unauthorized access.
- * Access control: Limits who can use certain data
- * Compliance: Follows laws like GDPR or POPIA.

Why it's a Problem: It leads to unfair discrimination and retaliatory actions incentives instead of making objective decisions.

6. Spreading misinformation using AI

Example: AI-generated fake news or deepfake videos that many people believe false information

Why it's a Problem: It can mislead the public, damage reputations, and even influence elections or social stability.

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32. Bias in AI refers to systematic and unfair differences in how an AI system treats people or makes decisions. It happens when an AI model produces results that are skewed in favor of or against certain groups (e.g., based on gender, race, age or location).

What it means for an AI model to be biased

An AI model is biased when:

- * It learns from data that is unbalanced or incomplete.
- * It reflects human prejudice or historical inequality in the training data.
- * It makes inconsistent or unfair predictions for different groups of people.

Real-world example of AI bias

A well known example is facial recognition system that performs less accurately on people with darker skin tones and on women compared to lighter-skinned men.

This happens because the training datasets contained more images of lighter-skinned individuals so the model learned patterns that didn't generalize well to everyone.

Potential consequences

- * Wrong identification or false matches (e.g., in security or police systems)

- * unfair treatment in hiring or lending decisions
- * Discrimination and loss of trust in AI systems
- * Legal and ethical issues for organization using the AI

33) ~~RE~~ Comparison of the four roles

Role	main focus
Data Analyst	RE Explain data to find insights and support decisions
Data Scientist	Building models and making predictions from data.
Data Engineer	Building and managing data pipelines and infrastructure.
RE AI Engineer	RE developing and deploying AI/ML models into applications.

34) Business intelligence (BI) is the process of collecting, analyzing and presenting data to help business make better decisions. It turns raw data into useful insights through reports,

Dashboards, and Visualizations

Popular BI Tools

* Power BI

Helps businesses create interactive dashboards and reports. It connects to different data source and shows trends (e.g. sales performance), making it easier for managers to spot problems and opportunities quickly.

* Tableau

Focuses on strong data visualization. It turns complex data into easy-to-understand charts and graphs, helping business see patterns and make informed decisions faster.

How BI helps business

- * Identifies trends and patterns
- * Supports faster, data-driven decision
- * Improves planning and performance monitoring

Section E Personal Reflection

35. ~~I find~~ I found Artificial intelligence interesting because I didn't know it is so broad. I thought it ended on chatGPT.

36. I found JSON difficult to understand so.

I used ChatGPT to explain it and making examples.

37. Data and AI I see them shaping my future in a positive way or for better e.g. most jobs now are being replaced by AI. So when you invest in AI it means you will be on a safer place.

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